

A constructive proof that Bézout does not imply elementary divisor

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Abstract

Hägg and Mörtberg [1], with GPT-6 Astra, constructed a Bézout domain that is not an elementary divisor domain; their proof is classical. We outline a constructive proof, based on a Lean formalisation available at <https://github.com/coquand/bezout-constructive>. It has three ingredients. Chain lifting along explicit polynomial charts replaces compactness. Smoothness is carried as Jacobian-minor certificates. Points and components of the maximal locus are explicit. Apart from known constructive results, the only choice principle used is unique choice.

This note, written by Claude, documents a constructive version of the proof in [1] (<https://arxiv.org/abs/2609.35229>) and its Lean formalisation. It has not been reviewed by Hägg and Mörtberg.

1 The statement

Theorem 1 (Hägg–Mörtberg with GPT-6 Astra [1]). *There is a Bézout domain $R \supset \mathbb{Q}[x, y]$ over which the matrix $M = \begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix}$ has no diagonal reduction.*

The ring $R = \bigcup_n A_n$ is an ascending union of smooth, finitely generated $\mathbb{Q}$ -domains with $A_0 = \mathbb{Q}[x, y]$. Each step principalises one coprime pair of A_n , using a weighted centre (Abramovich–Temkin–Włodarczyk), the extended Rees algebra and the forcing algebra [2] $J_B(c) = B[\sigma]/(\sum c_i \sigma_i - 1)$. The original obstruction is topological: compact real sets with monotone surjections keep the Möbius bundle on $x^2 + y^2 = 1$ nonorientable. The classical proof uses Noetherianity, maximal ideals, minimal primes and Krull dimension throughout.

2 Constructive ingredients

Chain lifting instead of compact sets. The topological argument is replaced by a lifting property for finite chains of real points along each stage. At every stage it is proved with explicit charts and composed along the tower. The analytic input is a Kantorovich step with explicit radius and a polynomial inverse function theorem whose radius is uniform in the base point. It uses only positive inverses and positive roots, with no real division and no Markov principle.

Smoothness as certificates. A stage $A = \mathbb{Q}[Y]/(G)$ carries a *certificate of dimension n* . This is a comaximal cover by Jacobian minors h_k such that on each $D(h_k)$ some m of the generators cut out A and their $m \times m$ minor is invertible. The dimension is $n = |Y| - m$ on every piece. Certificates propagate along the tower:

- the forcing algebra $J = B[X_0, \dots, X_r]/(\sum c_i X_i - 1)$: on $D(c_i)$ one has $J[1/c_i] \cong B[1/c_i][X_j : j \neq i]$, so the c_i themselves give the cover and the dimension becomes $n + r$. This replaces the split-conormal argument of [1, Lemma 3.1], and $B \rightarrow J$ is injective iff $\text{Ann}(c) = 0$ [3, Lemma II.2.14];
- the Rees algebra: the weighted charts give the pieces, and the dimension becomes $n + 1$.

On a piece, A_h has the standard presentation $\mathbb{Q}[Z][Y, t]/(F_{\text{rel}}, F_{\text{coord}} - Z, ht - 1)$, whose Jacobian is a unit, so A_h is standard étale over $\mathbb{Q}[Z]$. Formal smoothness at a point follows from Newton lifting. The formalisation proves the isomorphism by two explicit inverse maps, so no kernel is computed.

Points instead of maximal ideals. A *point* of A is a $\mathbb{Q}$ -algebra map $A \rightarrow L$ onto a finite extension of $\mathbb{Q}$. Its residue field is therefore discrete. Every local argument of the classical proof is run at a point:

- the invariant and its admissibility (decided at points);
- the maximal locus as an intersection over explicit points;
- primality of s in the Rees algebra;
- the drop of the invariant at the vertex.

Negative goals use $\neg\neg$ -stability of membership at a point. The local–global principle for comaximal localisations [3] assembles the pieces. The Nullstellensatz in Jacobson form produces points: for $\mathfrak{p}$ prime and $a \notin \mathfrak{p}$ there is a point that kills $\mathfrak{p}$ and not a .

Explicit inputs.

- Factorisation over $\mathbb{Q}$ uses Kronecker’s method and Gauss’s lemma. It transfers to $K(t)$ and to $K[X]/(p)$.
- Minimal primes follow Perdry’s strongly Noetherian approach [4].
- The invariant (order at a point) is the escape level of a Rees bar induction, so no unbounded search is needed. The weighted minimiser is a search over an explicit box.

Unique choice. The tower is a recursively defined sequence of rings, and each stage is selected by a property that determines it uniquely:

- the maximal invariant;
- the canonical component, the least canonical generator list in a decidable order coming from an explicit enumeration of the ring;
- the common denominator, the least admissible index, and the canonical generators of the next presentation.

Passing from $\exists!$ to data is unique choice. It is valid in Bishop-style mathematics. In Lean, however, it is an extra principle, because **Prop** does not eliminate into **Type**. Nothing stronger is used.

3 Status of the formalisation

- `main_theorem` and `main_theorem_tower` are proved in Lean 4/Mathlib with the standard axioms (9188 build jobs).
- The constructive layer passes an audit of 1921 declarations. It relies on 8 trusted primitives:
 - sign splitting and cotransitivity on $\mathbb{R}$;
 - positive inverses and positive roots;
 - the Banach fixed point with rate $\frac{1}{2}$;
 - the polynomial gcd over a discrete field;
 - the primitive-element shift (a bounded decidable search, justified by $\neg\neg$).
- In the dependency cone of the main theorem (4812 declarations), no project declaration uses classical logic directly. The standard axioms still appear, but only through Mathlib lemmas at the level of propositions (Hilbert basis, localisation, finite-set bookkeeping), which are known constructive facts. The tower is defined by well-founded recursion together with unique choice.

References

- [1] C. Hägg and A. Mörtberg, *A Bézout domain that is not an elementary divisor domain*, arXiv:2609.35229, 2026, <https://arxiv.org/abs/2609.35229>. The proof was found with GPT-6 Astra and formalised in Lean with Codex; <https://github.com/Zelaron/bezout-counterexample-lean>.
- [2] M. Hochster, *Solid closure*, Contemp. Math. 159 (1994), 103–172.
- [3] H. Lombardi and C. Quitté, *Commutative Algebra: Constructive Methods*, Springer, 2015.
- [4] H. Perdry, *Strongly Noetherian rings and constructive ideal theory*, J. Symbolic Comput. 37 (2004).